

NASA SDS Product Specification

Level-3 Soil Moisture

L3_SME2

Rev C (version 1.0.0)

JPL D-107677

November 8, 2024

Authors: Xiaodong Huang and Steven K. Chan

Copies of this document obtained outside of JPL's Product Lifecycle Management (PLM) tool may not be current. Please contact Heresh Fattahi (heresh.fattahi@jpl.nasa.gov) to determine if a more recent revision is available.

National Aeronautics and
Space Administration

Jet Propulsion Laboratory
California Institute of Technology
Pasadena, California

This document has been reviewed and determined not to contain export controlled technical data.

SIGNATURE PAGE

Prepared by:

Email approval on file 02-Jan-2025
Xiaodong Huang, NISAR Soil Moisture ADT Product Lead Date

Electronic signature on file 06-Jan-2025
Steven K. Chan, NISAR Soil Moisture ST/ADT Liaison Date

Approved by:

Electronic signature on file 03-Jan-2025
Ana Maria Guerrero, NISAR Mission System Manager Date

Electronic signature on file 02-Jan-2025
Paul Rosen, NISAR Project Scientist Date

Electronic signature on file 06-Jan-2025
Cecilia Cheng, NISAR SDS Manager Date

Electronic signature on file 07-Jan-2025
Heresh Fattahi, NISAR ADT Lead Date

Electronic signature on file 06-Jan-2025
Helen Mortensen, NISAR SDS System Engineer Date

This document has been reviewed and determined not to contain export controlled technical data.

EPDM ELECTRONIC SIGNATURES

User-Group/Role	...	Decision	...	Date
Zadeh, Sam (szadeh)-CM and IM/CME	...	Approve	...	25-Feb-2025 23:29
Cheng, Cecilia S (cecilia)-JPL Consumer/Proje...	...	Approve		06-Jan-2025 15:42
Chan, Steven K (chan)-JPL Consumer/Project ...	...	Approve		06-Jan-2025 10:08
Fattahi, Heresh (fattahi)-JPL Author/JPL Autho...	...	Approve		07-Jan-2025 11:20
Guerrero, Ana Maria P (ana)-JPL Author/JPL A...	...	Approve		03-Jan-2025 08:08
Mortensen, Helen B (hbmorten)-JPL Consume...	...	Approve		06-Jan-2025 09:03
Rosen, Paul A (parosen)-JPL Consumer/Projec...	...	Approve		02-Jan-2025 19:28

ACKNOWLEDGEMENT

The research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004).

© 2024 California Institute of Technology. Government sponsorship acknowledged.

This document has been reviewed and determined not to contain export controlled technical data.

DOCUMENT CHANGE LOG

Revision	Cover Date	Sections Changed	ECR #	Reason, ECR Title, LRS #*
Initial Release	Jan 15, 2023	All	N/A	New document LRR 069683
Release 3.3	Apr 28, 2023	All	N/A	For Release 3.3 URS CL# 23-2430
Initial (R3.4)	Oct 02, 2023	Section 2.2 and 4.2.	N/A	For Release 3.4
Rev A	Feb 07, 2024	Section 1.3, 2.2, 3.3, and 4.2.	N/A	For Release 4.0 Update the Table 2.1.1 Update the references Add the granuleID and soil moisture attributes to tables Update he captions of Tables and Figures Update the description of the HDF5 file, deleted Table 3.2.2 and update Table 3.2.3. URS CL#24-1293.
Rev B	May 30, 2024	Section 2.2 and 4.2	N/A	For Release 4.0.2 Update the Table 2.2.1 Update the Table 4.2.1. URS CL#24-3346
Rev C	Nov 08, 2024	Section 2.2, 3.2.1, 3.2.2, and 4.2	N/A	For Release R4.0.4 Update the Table 2.2.1 Update the Table 3.2.1 Update the Table 3.2.2 Update the Table 4.2.1. URS CL#24-6460

** Include the JPL Limited Release System (LRS) clearance number for each revision to be shared with foreign partners.*

This document has been reviewed and determined not to contain export controlled technical data.

TABLE OF CONTENTS

1 INTRODUCTION 7
1.1 Purpose of Description 7
1.2 Document Organization 7
1.3 Applicable and Reference Documents 7
2 PRODUCT OVERVIEW 9
2.1 Product Background 9
2.2 L3_SME2 Overview 11
2.3 L3_SME2 Filename Convention 11
3 PRODUCT ORGANIZATION 14
3.1 File Format 14
3.1.1 HDF5 File 14
3.1.2 HDF5 Group 14
3.1.3 HDF5 Dataset 14
3.1.4 HDF5 Datatype 14
3.1.5 HDF5 Attribute 16
3.2 File Organization 16
3.2.1 Groups 16
3.2.2 Field Level Metadata 16
3.2.3 Variable Metadata (HDF5 Attributes) 17
3.3 Granule Definition 18
3.4 File Naming Convention 18
3.5 Temporal Organization 18
3.6 Spatial Organization 18
3.7 Spatial Sampling and Resolution 18
4 LEVEL 3 SOIL MOISTURE PRODUCT 19
4.1 Dimensions and Shapes of Data 19
4.2 Product Specification 19
APPENDIX A: ACRONYMS 41

This document has been reviewed and determined not to contain export controlled technical data.

TABLE OF TABLES

TABLE 2.1.1 NISAR L0-L3 PRODUCTS BY LINEAGE 10
TABLE 2.1.2 NISAR L0-L3 PRODUCTS BY SCIENCE 11
TABLE 3.1.1 HDF5 ATOMIC DATATYPES 15
TABLE 3.1.2 NISAR HDF5 DERIVED AND COMPOUND DATATYPES 15
TABLE 3.2.1 GROUP ORGANIZATION AT THE TOP LEVEL OF A NISAR HDF5 FILE 16
TABLE 3.2.2 GLOBAL ATTRIBUTES IN SME2 PRODUCT 16
TABLE 3.2.3 COMMON VARIABLE ATTRIBUTES IN SME2 PRODUCT 17
TABLE 3.2.4 STATISTICAL ATTRIBUTES FOR REAL-VALUED HDF5 DATASETS 17
TABLE 3.2.5 L3 SME2 HDF5 DATASETS POPULATED WITH STATISTICAL ATTRIBUTES 17
TABLE 4.2.1 NISAR L3_SME2 OUTPUT FIELDS 19

TABLE OF FIGURES

Figure 2.1.1 Soil moisture retrieval coverage. White land areas: retrievals performed with quality flag raised only in case of time-varying properties (e.g., heavy precipitation). Light red areas (urban build-up, permanent snow and ice cover, or permanent inland waterbodies) will be flagged and no NISAR soil moisture retrievals will be performed. Gray areas (VWC > 5 kg/m^2) will be flagged but soil moisture retrievals will be performed. 9

This document has been reviewed and determined not to contain export controlled technical data.

1 INTRODUCTION

1.1 Purpose of Description

This document provides a specification of the NASA-ISRO Synthetic Aperture Radar (NISAR) L-SAR Level-3 Soil Moisture product to be generated by the NASA Science Data System (SDS) and provided to the NASA Alaska Satellite Facility Distributed Active Archive Center (ASF DAAC). This data product is referenced by the short name L3_SME2, where the numeral “2” signifies the fact that the product is on the 200-meter global Equal-Area Scalable Earth (EASE) Grid 2.0 projection [RD7][RD8].

1.2 Document Organization

Section 2 provides an overview of the product, including its purpose, and latency.

Section 3 provides the structure of the product, including granule definition, file organization, spatial resolution, temporal and spatial organization of the content, the size, and data volume.

Section 4 provides a detailed identification of the individual fields within the L3_SME2 product, including for example their units, size, and coordinates.

Appendix A provides a listing of the acronyms used in this document.

1.3 Applicable and Reference Documents

Applicable documents levy requirements on areas addressed in this document. Reference documents are cited to provide additional information to readers. In case of conflict between the applicable documents and this document, the Project shall review the conflict to find the most effective resolution.

Applicable Documents

[AD1] NISAR NASA SDS Level 4 Requirements, JPL D-95655, Rev A, February 6, 2024

[AD2] NISAR NASA SDS Algorithm Development Plan, JPL D-95678, Initial, September 12, 2019

[AD3] NISAR Science Data Management and Archive Plan, JPL D-80828, June 1, 2016

[AD4] NISAR Science Management Plan, JPL D-76340, Rev A, August 14, 2018

[AD5] NISAR SDS ADT Calibration and Validation Plan, JPL D-102256, Rev A, November 20, 2023

[AD6] NISAR NASA SDS L4 Software Management Plan (SMP), JPL D-95656, Rev A, September 19, 2019

[AD7] ISO-19115-2, https://www.iso.org/obp/ui/#iso:std:iso:19115:-2:ed-2:v1:en

This document has been reviewed and determined not to contain export controlled technical data.

Reference Documents

[RD1] NISAR L3 Soil Moisture Data Product Algorithm Theoretical Basis Document (ATBD), JPL D-107679, Release 3.3, April 28, 2023.

[RD2] EOSDIS Handbook, July 2016, retrieved from https://cdn.earthdata.nasa.gov/conduit/upload/5980/EOSDISHandbookWebFinaL2.pdf

[RD3] NISAR SDS L-SAR File Naming Conventions, JPL D-102255, Rev A, April 28, 2023

[RD4] NISAR L1_RSLC Product Specification Document, JPL D-102268, Rev D, November 8, 2024

[RD5] HDF5 documentation at https://portal.hdfgroup.org/display/HDF5/HDF5

[RD6] Eineder, M. (2003), Efficient simulation of SAR interferograms of large areas and of rugged terrain, IEEE Transactions on Geoscience and Remote Sensing, 41(6), 1415-1427.

[RD7] Brodzik, M. J., B. Billingsley, T. Haran, B. Raup, M. H. Savoie. 2012. EASE-Grid 2.0: Incremental but Significant Improvements for Earth-Gridded Data Sets. *ISPRS International Journal of Geo-Information*, 1(1):32-45, doi:10.3390/ijgi1010032.

[RD8] Brodzik, M. J., B. Billingsley, T. Haran, B. Raup, M. H. Savoie. 2014. Correction: Brodzik, M. J. et al. EASE-Grid 2.0: Incremental but Significant Improvements for Earth-Gridded Data Sets. *ISPRS International Journal of Geo-Information* 2012, 1, 32-45. ISPRS International Journal of Geo-Information, 3(3):1154-1156, doi:10.3390/ijgi3031154.

The NISAR Level 1 science requirements are translated into requirements on the various spacecraft and instrument systems, including the requirements related to the processing system producing the L0-L3 products. These SDS requirements [AD1] fall into three general categories: resolution requirements, radiometric and spatial location accuracy requirements, and latency and throughput requirements. Note that there is no geophysical retrieval accuracy imposed on the NISAR L3_SME2 product per agreement with the product sponsor – the Satellite Needs Working Group (SNWG); retrieval performance will be attempted on a best-effort basis.

This document has been reviewed and determined not to contain export controlled technical data.

2 PRODUCT OVERVIEW

2.1 Product Background

Soil moisture is a key variable that impacts the exchange of water and heat energy between the land surface and the atmosphere. It plays an important role in the development of weather patterns and precipitation. Soil moisture also strongly affects the amount of precipitation that runs off into nearby streams and rivers. Soil moisture information can be used for reservoir management, early warning of droughts, irrigation scheduling, and crop yield forecasting. Soil moisture data has the potential to significantly improve the accuracy of short-term weather forecasts and reduce the uncertainty of long-term projections of how climate change will impact Earth's water cycle [e.g., Entekhabi et al., 2010]. Satellite remote sensing of soil moisture has advanced significantly over the last decade due to the success of the Soil Moisture and Ocean Salinity (SMOS) [Kerr et al. 2010] and Soil Moisture Active Passive (SMAP) [Entekhabi et al., 2010] missions, both of which provide global soil moisture retrievals on an approximate 3-day revisit interval at an accuracy of approximately 0.04 m^3/m^3. A key limiting factor of SMAP and SMOS soil moisture measurements is their coarse spatial resolution (~40 km), which limits their utility for field-scale (i.e., ~200 m) agricultural monitoring. The unique capabilities of the NISAR mission [Rosen et al, 2015; Rosen et al., 2017] clearly motivate the production of field-scale (~200 m) land surface soil moisture products.

The salient characteristics of NISAR L3_SME2 are:

- NISAR will produce a global soil moisture product with a repeat interval of 12 days (6 days when using both ascending/descending modes)
- Latency: 72 hours
- Resolution: 200 m
- Accuracy goal: 0.06 m^3/m^3 (over non-urban areas with VWC below 5 kg/m^2, with no permanent snow and ice cover, and no permanent inland waterbodies as shown in Figure 2.1.1.

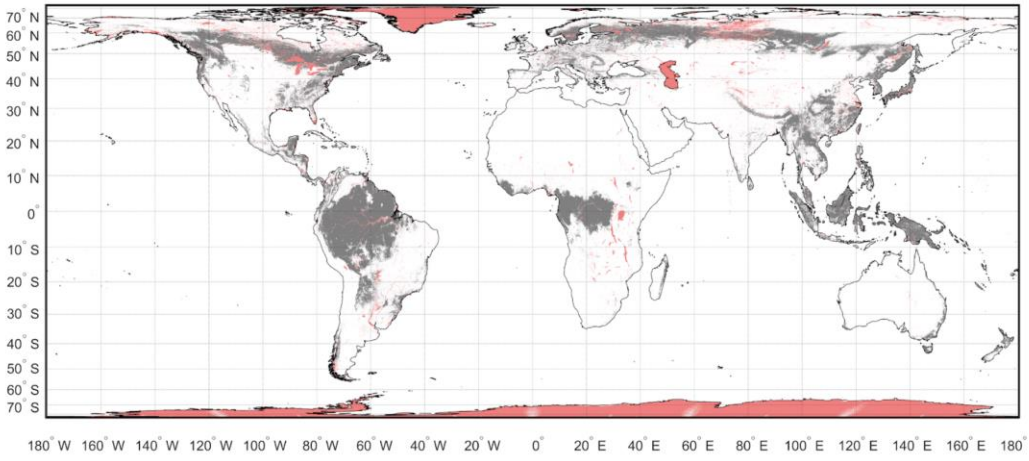


Figure 2.1.1 Soil moisture retrieval coverage. White land areas: retrievals performed with quality flag raised only in case of time-varying properties (e.g., heavy precipitation). Light red areas (urban build-up, permanent snow and ice cover, or permanent inland waterbodies) will be flagged and no NISAR soil moisture retrievals will be performed. Gray areas (VWC > 5 kg/m^2) will be flagged but soil moisture retrievals will be performed.

This document has been reviewed and determined not to contain export controlled technical data.

Table 2.1.1 NISAR L0-L3 products by lineage

Product	Scope	Description	Granularity
Radar Raw Science Telemetry (RRST)	Global	This L0A product is the raw downlinked data delivered to SDS	By downlinked files
Radar Raw Signal Data (RRSD)	Global	This L0B product is corrected, aligned radar pulse data derived from the RRST products and used for further processing	By radar observation, i.e., continuous data collected in a single radar mode
Calibration Raw Signal Data (CRSD)	Global	This L0B product contains instrument calibration data.	By radar datatake, i.e., a sequence of observations for one radar-on period

Product	Scope	Description	Granularity
Range-Doppler Single Look Complex (RSLC)	Global	The L1 RSLC product contains focused SAR images in range-Doppler coordinates. The RSLC is an input to other L1 or L2 products.	On pre-defined track/frame.
Range-Doppler Nearest-Time Interferogram (RIFG)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only	Multi-looked interferogram in Range Doppler coordinates, ellipsoid and topographic phase flattened and formed with precise coregistration using geometrical offsets and high-resolution pixel offsets obtained from incoherent cross correlation.	On pre-defined track/frame
Range-Doppler Nearest-Time Pixel Offsets (ROFF)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only.	Unfiltered and unculled layers of pixel offsets in Range Doppler coordinates with different resolutions obtained from incoherent cross correlation.	On pre-defined track/frame.
Range-Doppler Nearest-Time Unwrapped Interferogram (RUNW)	Antarctica, Greenland, and selected mountain glaciers. Nearest pair in time and co-pol channels only.	Multi-looked, ellipsoid and topography flattened unwrapped interferogram in Range Doppler coordinates.	On pre-defined track/frame

Product	Scope	Description	Granularity
Geocoded SLC (GSLC)	Global (Antarctica and Greenland excluded) and all channels.	Single Look Complex SAR image on geocoded map coordinate system.	On pre-defined track/frame
Geocoded Nearest-Time Pixel Offsets (GOFF)	Antarctica, Greenland, and mountain glaciers. Nearest pair in time and co-pol channels only.	Unfiltered and unculled layers of pixel offsets in with different resolutions obtained from incoherent cross correlation and geocoded on map coordinate system.	On predefined track/frame
Geocoded Nearest-Time Unwrapped Interferogram (GUNW)	Global. Nearest pair in time and co-pol channels only.	Geocoded, multi-looked, ellipsoid and topography flattened unwrapped interferogram.	On pre-defined track/frame

This document has been reviewed and determined not to contain export controlled technical data.

Product	Scope	Description	Granularity
Geocoded Polarimetric Covariance Matrix (GCOV)	Global and all channels. Single/Dual/Quad pol.	Geocoded, multi-looked polarimetric covariance matrix.	On pre-defined track/frame
Geocoded Soil Moisture on the 200-meter global EASE Grid 2.0 projection	Global; Snapshot algorithms require a single GCOV granule; time series algorithms require up to three GCOV granules	Soil moisture inferred from NISAR backscatter observations posted on the 200-meter global EASE Grid 2.0 projection. The resulting geographical extent is bounded by that of the L2 GCOV on UTM projection.	On pre-defined track/frame

Table 2.1.2 NISAR L0-L3 products by science

Product	Description
Level 0A	Unprocessed instrument data with some communications artifacts removed, but without reconstruction of missing data and reordering of samples from the instrument. May still contain bit errors and missing data that needs reconstruction.
Level 0B	Reconstructed, unprocessed instrument data at original resolution, time ordered, all communications artifacts removed.
Level 1	Processed instrument data, focused to full resolution complex images, time referenced and annotated with ancillary information, including radiometric and relevant geometric calibration coefficients and georeferencing parameters (i.e. platform ephemeris) computed and appended, in natural radar coordinates.
Level 2 Category 1	Derived radar-specific parameters at the same or reduced resolution as Level 1 imagery, but resampled and geocoded to a geographic or ellipsoidal grid.
Level 2 Category 2	Derived radar-specific parameters at reduced resolution, in original Level 1 coordinates.
Level 3	Geophysical parameters derived from Level 2 data that have been spatially and/or temporally re-sampled to a global grid.

2.2 L3_SME2 Overview

NISAR backscatter observations will be used to estimate a global high resolution soil moisture product (200 m). This product will be provided on average twice every 12 days. The NISAR soil moisture product is expected to have a data latency of 72 hours (3 days). The NISAR L2 backscatter product has a data latency of 48 hours. Soil moisture estimates will be provided over areas with dense vegetation (VWC greater than 5 kg/m^2) but will be flagged during the retrieval process. Areas with urban build-up, permanent snow and ice cover, and permanent inland waterbodies (shown in red) will be flagged and no soil moisture retrieval will be performed. No soil moisture retrieval will be performed over areas with excessive precipitation, frozen ground, or areas with snow on ground. The NISAR soil moisture will have an accuracy goal of 0.06 m^3/m^3 over unflagged areas with vegetation water content below 5 kg/m^2. The 5 + 5 MHz product will be aggregated to 400 meters, but is still on the 200m EASE Grid.

2.3 L3_SME2 Filename Convention

The following syntax governs the filename convention for NISAR L3_SME2 granules available from the ASF DAAC:

This document has been reviewed and determined not to contain export controlled technical data.

NISAR_*IL_PT_PROD_CYL_REL_P_FRM_MODE_POLE_S_StartDateTime_EndDateTime_CRID_A_C_LOC_CTR*.EXT

where

- NISAR – 5 char for mission: **NISAR**
- *I* – 1 char for Instrument: **L** for L-SAR, **S** for S-SAR
- *L* – 1 char for Level: **1** or **2** or **3**
- *PT* – 2 char for Processing Type:
  - **PR –** Production
  - **UR –** Urgent Response
- *PROD* – 4 chars for Product Identifier: **RSLC, GSLC, GCOV, SME2**
- *CYL* – 3 chars for CYcLe number in the mission, each cycle represents 12 days, zero padded, starting at 001.
- *REL* – 3 chars for RELative orbit track number within a cycle, resets to 1 with a cycle number increment, zero padded.Valid values: 001-173.
- *P* – 1 char for direction of movement of the satellite at the time of imaging]:
  - **A** for Ascending
  - **D** for Descending
- *FRM* – 3 chars for track frame number, a segment of an orbital track corresponding to the product, zero padded Valid Values: 001-176 on each track. See sec 4.5 [RD3]
- *MODE* – 4 chars for Bandwidth Mode Code of Primary and Secondary Bands: **40, 20, 77, 05, or 00** (only if the secondary band is missing)
- *POLE* – 4 chars for Polarization of the data for the primary and secondary bands. Each band uses a two character code among the following:
  - **SH** = HH – Single Polarity (H transmit and receive)
  - **SV** = VV – Single Polarity (V transmit and receive)
  - **DH** = HH/HV – Dual Polarity (H transmit)
  - **DV** = VV/VH – Dual Polarity (V transmit)
  - **CL**= LH/LV – Compact Polarity (Left transmit)
  - **CR** = RH/RV – Compact Polarity (Right transmit)
  - **QP** = HH/HV/VV/VH – Quad Polarity
  - **NA** if band does not exist

  For example, a “quasi-quad” polarization mode would be noted **DHDV** while a “quasi-dual” polarization mode would be noted **SHSV**.
- **S – 1 char for source of data for the product**
  - **A = Acquired source of the observation, single mode**
  - **M = Mixed source of observations, mixed mode**
- *StartDateTime* – 15 chars for Radar Start Time of the data processed as zero Doppler contained in the file as YYYYMMDDTHHMMSS, UTC
- *EndDateTime* – 15 chars for Radar End Time of the data processed as zero Doppler contained in the file as YYYYMMDDTHHMMSS, UTC
- *CRID* – 6 chars for Composite Release Identifier

This document has been reviewed and determined not to contain export controlled technical data.

    - Format of EPMMmm (See section 4.1 [RD3])

- *A* – 1 char for Product Accuracy or Fidelity of the Orbit Ephemeris and Radar Pointing:
  • **P, M, N,** or **F**, See section 4.2 [RD3]
- *C* – 1 char as Coverage Indicator: **F** for Full or **P** for Partial. See section 4.3 [RD3]

- *LOC* – 1 char to represent the location of the Science Data System. J for JPL, recommends N for NRSC. See section 4.6 [RD3]

- *CTR* – 3 chars for Product Version. See section 4.4 [RD3]

- *EXT* – 1 to n chars for Extension: **h5, met, log**

NOTES:
- 17 spacing characters: 16 for underscores plus one period
- Total number of characters: **90** excluding extension

Example:
NISAR_L3_PR_SME2_001_005_A_219_4020_DHDV_M_20220104T182346_20220104T183426_P01101_M_P_J_001.h5

This document has been reviewed and determined not to contain export controlled technical data.

3 PRODUCT ORGANIZATION

3.1 File Format

All NISAR standard products are in the Hierarchical Data Format version 5 (HDF5, [RD5]). HDF5 is a general-purpose file format and programming library for storing scientific data. The National Center for Supercomputing Applications (NCSA) at the University of Illinois developed HDF to help scientists share data more easily. Use of the HDF library enables users to read HDF files regardless of the underlying computing environments. HDF files are equally accessible in Fortran, C/C++, and other high-level computation packages such as IDL, MATLAB, and python.

The HDF Group, a spin-off organization of the NCSA, is responsible for development and maintenance of HDF. Users should reference The HDF Group website at https://portal.hdfgroup.org/display/HDF5/HDF5 [RD5] to download HDF software and documentation.

HDF5 represents a significant departure from the conventions of previous versions of HDF. The changes that appear in HDF5 provide flexibility to overcome many of the limitations of previous releases. The basic building blocks have been largely redefined, and are more powerful but less numerous. The key concepts of the HDF5 Abstract Data Model are Files, Groups, Datasets, Datatypes, Attributes and Property Lists. The following sections provide a brief description of each of these key HDF5 concepts.

3.1.1 HDF5 File

A File is the abstract representation of a physical data file. Files are containers for HDF5 Objects. These Objects include Groups, Datasets, and Datatypes.

3.1.2 HDF5 Group

Groups provide a means to organize the HDF5 Objects in HDF5 Files. Groups are containers for other Objects, including Datasets, named Datatypes and other Groups. In that sense, groups are analogous to directories that are used to categorize and classify files in standard operating systems.

The notation for files is identical to the notation used for Unix directories. The root Group is “/”. A Group contained in root might be called “/myGroup.” Like Unix directories, Objects appear in Groups through “links”. Thus, the same Object can simultaneously be in multiple Groups.

3.1.3 HDF5 Dataset

The Dataset is the HDF5 component that stores user data. Each Dataset associates with a Dataspace that describes the data dimensions, as well as a Datatype that describes the basic unit of storage element. A Dataset can also have Attributes.

3.1.4 HDF5 Datatype

A Datatype describes a unit of data storage for Datasets and Attributes. Datatypes are subdivided into Atomic and Composite Types.

Atomic Datatypes are analogous to simple basic types in most programming languages. HDF5 Atomic Datatypes include Time, Bitfield, String, Reference, Opaque, Integer, and Float. Each atomic type has a specific set of properties. Examples of the properties associated with Atomic Datatypes are:

This document has been reviewed and determined not to contain export controlled technical data.

- Integers are assigned size, precision, offset, pad byte order, and are designated as signed or unsigned.
- Strings can be fixed or variable length, and may or may not be null-terminated.
- References are constructed within HDF5 Files that point to other HDF5 Objects in the same file.

HDF5 provides a large set of predefined Atomic Datatypes. Table 3.1.1 lists the Atomic Datatypes that are used in NISAR data products.

Table 3.1.1 HDF5 Atomic Datatypes

HDF5 Atomic Datatypes	Description
H5T_STD_U8LE	8-bit little-endian unsigned integer
H5T_STD_U16LE	16-bit little-endian unsigned integer
H5T_STD_U32LE	32-bit little-endian unsigned integer
H5T_STD_U64LE	64-bit little-endian unsigned integer
H5T_STD_I8LE	8-bit little-endian signed integer
H5T_STD_I16LE	16-bit little-endian signed integer
H5T_STD_I32LE	32-bit little-endian signed integer
H5T_STD_I64LE	64-bit little-endian signed integer
H5T_IEEE_F32LE	32-bit little-endian floating point
H5T_IEEE_F64LE	64-bit little-endian floating point
H5T_C_S1	character string made up of one or more bytes

Derived Datatypes are user-defined variants of predefined Atomic Datatypes where the data organization has been modified at the bit-level. Derived data types are particularly useful for representing custom N-bit integers and floating-point numbers.

Composite Datatypes incorporate sets of Atomic datatypes. Composite Datatypes include Array, Enumeration, Variable Length and Compound.

- The Array Datatype defines a multi-dimensional array that can be accessed atomically.
- Variable Length presents a 1-D array element of variable length. Variable Length Datatypes are useful as building blocks of ragged arrays.
- Compound Datatypes are composed of named fields, each of which may be dissimilar Datatypes. Compound Datatypes are conceptually equivalent to structures in the C programming language.

Named Datatypes are explicitly stored as Objects within an HDF5 File. Named Datatypes provide a means to share Datatypes among Objects. Datatypes that are not explicitly stored as Named Datatypes are stored implicitly. They are stored separately for each Dataset or Attribute they describe.

NISAR products employ the following Derived and Compound Datatypes.

Table 3.1.2 NISAR HDF5 Derived and Compound Datatypes

This document has been reviewed and determined not to contain export controlled technical data.

Description	Comments
16-bit little-endian floating point	“binary16” half precision type in IEEE 754-2008 standard. Matches numpy.float16 type in Python. We will refer to this type as H5T_IEEE_F16LE or Float16 in our documents.

H5T_COMPOUND
{
16-bit little-endian floating-point "r";
16-bit little-endian floating-point "i";
}
	Complex numbers made up of two half precision floating point numbers. We will refer to this type as H5T_CPX_F16LE or CFloat16 in our documents.

H5T_COMPOUND
{
32-bit little-endian floating-point "r";
32-bit little-endian floating-point "i";
}
	Complex numbers made of two single precision floating point numbers. We will refer to this type as H5T_CPX_F32LE or CFloat32 in our documents.

H5T_COMPOUND
{
64-bit little-endian floating-point "r";
64-bit little-endian floating-point "i";
}
	Complex numbers made of two double precision floating point numbers. We will refer to this type as H5T_CPX_F64LE or CFloat64 in our documents.

3.1.5 HDF5 Attribute

An Attribute is a small aggregate of data that describes Groups or Datasets. Like Datasets, Attributes are also associated with a particular Dataspace and Datatype. Attributes cannot be subsetted or extended. Attributes themselves cannot have Attributes.

3.2 File Organization

3.2.1 Groups

All NISAR HDF5 files are organized as groups with no actual data at the root (“/”) level. Table 3.2.1.1 shows the general layout of the HDF5 files that are generated by the NISAR Science Data System. All data are organized under “/science/LSAR/” with data from the L-SAR instrument.

Table 3.2.1 Group organization at the top level of a NISAR HDF5 File

Group Name	Description
/science/LSAR/	All science data from the L-SAR instrument is organized under this group
/science/LSAR/identification/	File level metadata for cataloging, archiving the particular granule

3.2.2 Field Level Metadata

Global metadata at the file level are currently given as Global attributes shown in Table 3.2.2.

Table 3.2.2 Global attributes in SME2 product

Attribute	Format	Description
Conventions	string	CF-1.7
title	string	NISAR L3_SME2 Product

This document has been reviewed and determined not to contain export controlled technical data.

institution	string	NASA JPL
mission_name	string	NISAR
reference_document	string	D-107677 NISAR NASA SDS Product Specification Level-3 Soil Moisture
contact	string	nisar-sds-ops@jpl.nasa.gov

3.2.3 Variable Metadata (HDF5 Attributes)

NISAR standards incorporate additional metadata that describe each HDF5 Dataset within the HDF5 file. Each of these metadata elements appears in an HDF5 Attribute that is directly associated with the HDF5 Dataset. Table 3.2.3 lists the HDF5 Attributes that soil moisture products typically employ.

Table 3.2.3 Common variable attributes in SME2 product

Attribute	Description
_FillValue	The value used to represent missing or undefined data.
_ChunkSizes	The size of the data blocks into which a dataset is divided for storage
grid_mapping	The spatial referencing information of a dataset
description	Miscellaneous information about the data or the methods to generate it.
long_name	Long descriptive variable name that indicates its content.
standard_name	Unique name used to identify the variable. A standard name contains no whitespace and is case sensitive
units	Units of the variable's data elements.
valid_max	Maximum theoretical value of the variable (not necessarily the same as maximum value of actual data)
valid_min	Minimum theoretical value of the (not necessarily the same as minimum value of actual data)

Some HDF5 Datasets are populated with statistical attributes. Table 3.2.4 describes statistical attributes added to real-valued HDF5 Datasets. The list of real-valued HDF5 Datasets for the standard L3_SME2 product is given in Table 3.2.5.

Table 3.2.4 Statistical attributes for real-valued HDF5 Datasets

Attribute	Description
min_value	Minimum value of a real-valued HDF5 Dataset
mean_value	Mean value of a real-valued HDF5 Dataset
max_value	Maximum value of a real-valued HDF5 Dataset
sample_stddev	Sample standard deviation of a real-valued HDF5 Dataset

Table 3.2.5 L3 SME2 HDF5 Datasets populated with statistical attributes

HDF5 Group Path	HDF5 Datasets
/science/LSAR/SME2/grids/	soilMoisture, soilMoistureUncertainty
/science/LSAR/SME2/grids/radarData/frequency[A|B]]/	noiseEquivalentBackscatter[HH|HV|VH|VV], numberOfLooks[HH|HV|VH|VV], sigma0[HH|HV|VH|VV]
/science/LSAR/SME2/grids/ancillaryData/	localIncidenceAngle, localIncidenceAngleUncertainty, waterbodyFraction,
/science/LSAR/SME2/grids/algorithmCandidates/DSG/	algorithmParameterBeta, algorithmParameterGamma, soilMoisture, soilMoistureUncertainty
/science/LSAR/SME2/grids/algorithmCandidates/TSR/	algorithmParameterAlpha1, algorithmParameterAlpha1Uncertainty,

This document has been reviewed and determined not to contain export controlled technical data.

	algorithmParameterAlpha2, algorithmParameterAlpha2Uncertainty , soilMoisture, soilMoistureUncertainty
/science/LSAR/SME2/grids/algorithmCandidates/PMI/	dielectricConstant, roughness, vegetationWaterContentHV, vegetationWaterContentNDVI, vegetationWaterContentRetrieved, soilMoisture, soilMoistureUncertainty

3.3 Granule Definition

NISAR L3_SME2 granules will be posted on the 200-meter global Equal-Area Scalable Earth (EASE) Grid 2.0 projection [RD7][RD8] and are expected to have a ground footprint of about 240 km x 240 km.

3.4 File Naming Convention

NISAR L3_SME2 Granule names will conform to the Standard Product File Naming Scheme [RD3]. See Section 2.

3.5 Temporal Organization

Temporal organization is not specifically applicable to the L3_SME2 product, although it is generally arranged in order of increasing azimuth time.

3.6 Spatial Organization

The L3_SME2 product is arranged on an equal-area Earth-fixed grid. Each grid cell is referenced by its zero-based row and column indices. Decreasing North is denoted by monotonic increasing row indices in the row direction, whereas increasing East is denoted by monotonic increasing column indices in the column direction.

3.7 Spatial Sampling and Resolution

Some salient characteristics of the output grid for the L3_SME2 product are:

1. (0,0) denotes the grid cell *center* of the northernmost-westernmost grid cell
2. (-0.5,-0.5) denotes the northernmost-westernmost grid *corner* of grid cell (0,0)
3. There are 73,080 rows and 173,520 columns in the global 200-m EASE Grid 2.0 projection on which L3_SME2 is posted.

This document has been reviewed and determined not to contain export controlled technical data.

4 LEVEL 3 SOIL MOISTURE PRODUCT

4.1 Dimensions and Shapes of Data

To simplify the description of the layout of data within the HDF5 file, a table of dimensions and shapes is used to represent the relationship between similarly sized datasets. This table is meant to be a guide to illustrate the shapes of the datasets in L3_SME2.

4.2 Product Specification

Table 4.2.1 NISAR L3_SME2 output fields.

Notes: 1. the string data type is a fixed-length byte string; 2. all timestamps are in UTC.

Identification Group		
science/LSAR/identification/absoluteOrbitNumber		
Type: UInt32		Shape: N/A
Description: Absolute orbit number		
/science/LSAR/identification/boundingPolygon		
Type: String		Shape: N/A
Description: OGR compatible WKT representing the bounding polygon of the image. Horizontal coordinates are WGS84 longitude followed by latitude (both in degrees), and the vertical coordinate is the height above the WGS84 ellipsoid in meters. The first point corresponds to the start-time, near-range radar coordinate, and the perimeter is traversed in counterclockwise order on the map. This means the traversal order in radar coordinates differs for left-looking and right-looking sensors. The polygon includes the four corners of the radar grid, with equal numbers of points distributed evenly in radar coordinates along each edge.		
	epsg	4326
	ogr_geometry	polygon
/science/LSAR/identification/diagnosticModeFlag		
Type: UInt8		Shape: N/A
Description: Indicates if the radar operation mode is a diagnostic mode (1-2) or DBFed science (0): 0, 1, or 2		
/science/LSAR/identification/frameNumber		
Type: UInt16		Shape: N/A
Description: Frame number		
/science/LSAR/identification/granuleId		
Type: String		Shape: N/A
Description: Unique granule identification name		
/science/LSAR/identification/instrumentName		
Type: String		Shape: N/A
Description: Name of the instrument used to collect the remote sensing data provided in this product		
/science/LSAR/identification/isDithered		
Type: String		Shape: N/A
Description: "True" if the pulse timing was varied (dithered) during acquisition, "False" otherwise.		
/science/LSAR/identification/isFullFrame		
Type: String		Shape: N/A

This document has been reviewed and determined not to contain export controlled technical data.

Description: "True" if the product fully covers a NISAR frame, "False" if partial coverage.		
	frameCoveragePercentage	Percentage of NISAR frame containing processed data
	thresholdPercentage	Threshold percentage used to determine if the product is full frame or partial frame
/science/LSAR/identification/isGeocoded		
Type: String		Shape: N/A
Description: Flag to indicate if the product data is in the radar geometry ("False") or in the map geometry ("True").		
/science/LSAR/identification/isJointObservation		
Type: String		Shape: N/A
Description: "True" if any portion of this product was acquired in a joint observation mode (e.g., L-band and S-band simultaneously), "False" otherwise		
/science/LSAR/identification/isMixedMode		
Type: String		Shape: N/A
Description: "True" if this product is a composite of data collected in multiple radar modes, "False" otherwise.		
/science/LSAR/identification/isUrgentObservation		
Type: String		Shape: N/A
Description: Flag indicating if observation is nominal ("False") or urgent ("True")		
/science/LSAR/identification/lookDirection		
Type: String		Shape: N/A
Description: Look direction, either "Left" or "Right"		
/science/LSAR/identification/missionId		
Type: String		Shape: N/A
Description: Mission identifier		
/science/LSAR/identification/orbitPassDirection		
Type: String		Shape: N/A
Description: Orbit direction, either "Ascending" or "Descending"		
/science/LSAR/identification/plannedDatatakeId		
Type: List of string		Shape: N/A
Description: List of planned datatakes included in the product		
/science/LSAR/identification/plannedObservationId		
Type: List of string		Shape: N/A
Description: List of planned observations included in the product		
/science/LSAR/identification/platformName		
Type: String		Shape: N/A
Description: Name of the platform used to collect the remote sensing data provided in this product		
/science/LSAR/identification/processingCenter		
Type: String		Shape: N/A
Description: Data processing center		
/science/LSAR/identification/processingDateTime		
Type: String		Shape: N/A

This document has been reviewed and determined not to contain export controlled technical data.

Description: Processing UTC date and time in the format YYYY-mm-ddTHH:MM:SS		
/science/LSAR/identification/processingType		
Type: String		Shape: N/A
Description: Nominal (or) Urgent (or) Custom (or) Undefined		
/science/LSAR/identification/productDoi		
Type: String		Shape: N/A
Description: Digital Object Identifier (DOI) for the product		
/science/LSAR/identification/productLevel		
Type: String		Shape: N/A
Description: Product level		
/science/LSAR/identification/productSpecificationVersion		
Type: String		Shape: N/A
Description: Product specification version which represents the schema of this product		
/science/LSAR/identification/productType		
Type: String		Shape: N/A
Description: Product type		
/science/LSAR/identification/productVersion		
Type: String		Shape: N/A
Description: Product version which represents the structure of the product and the science content governed by the algorithm, input data, and processing parameters		
/science/LSAR/identification/radarBand		
Type: String		Shape: N/A
Description: Acquired frequency band, either "L" or "S"		
/science/LSAR/identification/trackNumber		
Type: UInt32		Shape: N/A
Description: Track number		
/identification/zeroDopplerEndTime		
Type: String		Shape: N/A
Description: Azimuth end time of the product		
/identification/zeroDopplerStartTime		
Type: String		Shape: N/A
Description: Azimuth start time of the product		
/science/LSAR/SME2/grids/algorithmCandidates		
DSG Product Variables		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/algorithmParameterBeta		
Type: Float32		Shape: (productWidth, productLength)
Description: Algorithm sensitivity parameter between soil moisture and SAR co-pol backscatter		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value

This document has been reviewed and determined not to contain export controlled technical data.

	sample_stddev	standard deviation of the dataset
	long_name	Algorithm parameter beta
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/decibel/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/algorithmParameterGamma		
Type: Float32		Shape: (productWidth, productLength)
Description: Algorithm heterogeneity parameter as described by the correlation between SAR cross-pol and co-pol backscatters		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Algorithm parameter gamma
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/projection		
Type: Int32		
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	false_northing	0.0
	latitude_of_projection_origin	0.0
	longitude_of_projection_origin	0.0
	false_easting	0.0
	semi_major_axis	6378137
	inverse_flattening	298.25723
	longitude_of_central_meridian	0.0
	standard_parallel	30.0
	epsg_code	6933
	ellipsoid	WGS84
	grid_mapping_name	lambert_cylindrical_equal_area
	spatial_ref	PROJCS[WGS 84 / NSIDC EASE-Grid 2.0 Global,GEOGCS[WGS 84,DATUM[WGS_1984,SPHEROID[WGS 84,6378137,298.257223563,AUTHORITY[EPSG,7030]],AUTHORITY[EPSG,6326]],PRIMEM[Greenwich,0,AUTHORITY[EPSG,8901]],UNIT[degree,0.0174532925199433,AUTHORITY[EPSG,9122]],AUTHORITY[EPSG,4326]],PROJECTION[Cylindrical_Equal_Area],PARAMETER[standard_parallel_1,30],PARAMETER[central_meridian,0],PARAMETER[false_easting,0],PARAMETER[false_northing,0],UNIT[metre,1,AUTHORITY[EPSG,9001]],AXIS[Easting,EAST],AXIS[Northing,NORTH],AUTHORITY[EPSG,6933]]

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/SME2/grids/algorithmCandidates/DSG/retrievalQualityFlag		
Type: Int16		Shape: (productWidth, productLength)
Description: DSG retrieval quality flag while 0 means clear and non-zero means anomaly Flag bits 0: Retrieval recommended flag; 1: Retrieval attempted flag; 2: Retrieval success flag; 3: Soil moisture thresholds between (0.02 and 0.60); 4: NISAR σ_{hh} anomaly flag; 5: NISAR σ_{hv} anomaly flag; 6: NISAR σ_{vv} anomaly flag; 7: NISAR σ_{vh} anomaly flag		
	long_name	Retrieval quality flag
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/soilMoisture		
Type: Float32		Shape: (productWidth, productLength)
Description: DSG soil moisture		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/soilMoistureUncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: DSG soil moisture uncertainty		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture uncertainty
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/xCoordinates		
Type: Float64		Shape: productLength
Description: X coordinates in specified projection		
	standard_name	projection_x_coordinate
	long_name	x coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/xCoordinateSpacing		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float64		Shape: NA
Description: X coordinates spacing		
	long_name	X coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/yCoordinates		
Type: Float64		Shape: productWidth
Description: Y coordinates in specified projection		
	standard_name	projection_y_coordinate
	long_name	y coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/DSG/yCoordinateSpacing		
Type: Float64		Shape: NA
Description: Y coordinates spacing		
	long_name	Y coordinate of spacing
Unit: meters		
PMI Product Variables		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/cropType		
Type: Int8		Shape: (productWidth, productLength)
Description: PMI crop type 0: Wheat; 1: Rice; 2: Corn; 3: Soybean; 255: Nodata		
	long_name	Crop type
	_FillValue	-127
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/dielectricConstant		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI dielectric constant		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	PMI retrieved real part dielectric constant
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/projection		
Type: Int32		
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	false_northing	0.0

This document has been reviewed and determined not to contain export controlled technical data.

	latitude_of_projection_origin	0.0
	longitude_of_projection_origin	0.0
	false_easting	0.0
	semi_major_axis	6378137
	inverse_flattening	298.25723
	longitude_of_central_meridian	0.0
	standard_parallel	30.0
	epsg_code	6933
	ellipsoid	WGS84
	grid_mapping_name	lambert_cylindrical_equal_area
	spatial_ref	PROJCS[WGS 84 / NSIDC EASE-Grid 2.0 Global,GEOGCS[WGS 84,DATUM[WGS_1984,SPHEROID[WGS 84,6378137,298.257223563,AUTHORITY[EPSG,7030]],AUTHORITY[EPSG,6326]],PRIMEM[Greenwich,0,AUTHORITY[EPSG,8901]],UNIT[degree,0.0174532925199433,AUTHORITY[EPSG,9122]],AUTHORITY[EPSG,4326]],PROJECTION[Cylindrical_Equal_Area],PARAMETER[standard_parallel_1,30],PARAMETER[central_meridian,0],PARAMETER[false_easting,0],PARAMETER[false_northing,0],UNIT[metre,1,AUTHORITY[EPSG,9001]],AXIS[Easting,EAST],AXIS[Northing,NORTH],AUTHORITY[EPSG,6933]]
/science/LSAR/SME2/grids/algorithmCandidates/PMI/retrievalQualityFlag		
Type: Int16		Shape: (productWidth, productLength)
Description: PMI retrieval quality flag Flag bits 0: Retrieval recommended flag; 1: Retrieval attempted flag; 2: Retrieval success flag; 3: Soil moisture thresholds between (0.02 and 0.60); 4: NISAR σ_{hh} anomaly flag; 5: NISAR σ_{hv} anomaly flag; 6: NISAR σ_{vv} anomaly flag; 7: NISAR σ_{vh} anomaly flag		
	long_name	PMI retrieval quality flag while 0 means clear and non-zero means anomaly
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/roughness		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI surface roughness		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Surface roughness
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meters		

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/SME2/grids/algorithmCandidates/PMI/soilMoisture		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI soil moisture		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/soilMoistureUncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI soil moisture uncertainty		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture uncertainty
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/vegetationWaterContentHV		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI vegetation water content estimated from HV channel		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Vegetation water content HV
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: kilogram/meter^2		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/vegetationWaterContentNDVI		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI vegetation water content estimated from NDVI		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value

This document has been reviewed and determined not to contain export controlled technical data.

	sample_stddev	standard deviation of the dataset
	long_name	Vegetation water content NDVI
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: kilogram/meter^2		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/vegetationWaterContentRetrieved		
Type: Float32		Shape: (productWidth, productLength)
Description: PMI retrieved vegetation water content		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Vegetation water content retrieved
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: kilogram/meter^2		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/xCoordinates		
Type: Float64		Shape: productLength
Description: X coordinates in specified projection		
	standard_name	projection_x_coordinate
	long_name	x coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/xCoordinateSpacing		
Type: Float64		Shape: NA
Description: X coordinates spacing		
	long_name	X coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/yCoordinates		
Type: Float64		Shape: productWidth
Description: Y coordinates in specified projection		
	standard_name	projection_y_coordinate
	long_name	y coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/PMI/yCoordinateSpacing		
Type: Float64		Shape: NA
Description: Y coordinates spacing		
	long_name	Y coordinate of spacing
Unit: meters		
TSR Product Variables		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/algorithmParameterAlpha1		

This document has been reviewed and determined not to contain export controlled technical data.

Type: Float32		Shape: (productWidth, productLength)
Description: Retrieved soil moisture for previous time series (T-12)		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Algorithm parameter alpha1
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
	_FillValue	-9999
	grid_mapping	projection
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/algorithmParameterAlpha1Uncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: Retrieved soil moisture for previous time series (T-12)		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Algorithm parameter alpha1 uncertainty
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
	_FillValue	-9999
	grid_mapping	projection
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/algorithmParameterAlpha2		
Type: Float32		Shape: (productWidth, productLength)
Description: Retrieved soil moisture for previous time series (T-24)		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Algorithm parameter alpha2
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
	_FillValue	-9999
	grid_mapping	projection
Unit: meter^3/meter^3		

This document has been reviewed and determined not to contain export controlled technical data.

/science/LSAR/SME2/grids/algorithmCandidates/TSR/algorithmParameterAlpha2Uncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: Retrieved soil moisture for previous time series (T-24)		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Algorithm parameter alpha2 uncertainty
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
	_FillValue	-9999
	grid_mapping	projection
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/projection		
Type: Int32		
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	false_northing	0.0
	latitude_of_projection_origin	0.0
	longitude_of_projection_origin	0.0
	false_easting	0.0
	semi_major_axis	6378137
	inverse_flattening	298.25723
	longitude_of_central_meridian	0.0
	standard_parallel	30.0
	epsg_code	6933
	ellipsoid	WGS84
	grid_mapping_name	lambert_cylindrical_equal_area
	spatial_ref	PROJCS[WGS 84 / NSIDC EASE-Grid 2.0 Global,GEOGCS[WGS 84,DATUM[WGS_1984,SPHEROID[WGS 84,6378137,298.257223563,AUTHORITY[EPSG,7030]],AUTHORITY[EPSG,6326]],PRIMEM[Greenwich,0,AUTHORITY[EPSG,8901]],UNIT[degree,0.0174532925199433,AUTHORITY[EPSG,9122]],AUTHORITY[EPSG,4326]],PROJECTION[Cylindrical_Equal_Area],PARAMETER[standard_parallel_1,30],PARAMETER[central_meridian,0],PARAMETER[false_easting,0],PARAMETER[false_northing,0],UNIT[metre,1,AUTHORITY[EPSG,9001]],AXIS[Easting,EAST],AXIS[Northing,NORTH],AUTHORITY[EPSG,6933]]
/science/LSAR/SME2/grids/algorithmCandidates/TSR/retrievalQualityFlag		
Type: Int16		Shape: (productWidth, productLength)
Description: TSR retrieval quality flag while 0 means clear and non-zero means anomaly		

This document has been reviewed and determined not to contain export controlled technical data.

Flag bits 0: Retrieval recommended flag; 1: Retrieval attempted flag; 2: Retrieval success flag; 3: Soil moisture thresholds between (0.02 and 0.60); 4: NISAR σ_{hh} anomaly flag; 5: NISAR σ_{hv} anomaly flag; 6: NISAR σ_{vv} anomaly flag; 7: NISAR σ_{vh} anomaly flag		
	long_name	Retrieval quality flag
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/soilMoisture		
Type: Float32		Shape: (productWidth, productLength)
Description: TSR soil moisture		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/soilMoistureUncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: TSR soil moisture uncertainty		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture uncertainty
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/xCoordinates		
Type: Float64		Shape: productLength
Description: X coordinates in specified projection		
	standard_name	projection_x_coordinate
	long_name	x coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/xCoordinateSpacing		
Type: Float64		Shape: NA
Description: X coordinates spacing		
	long_name	X coordinate of spacing

This document has been reviewed and determined not to contain export controlled technical data.

Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/yCoordinates		
Type: Float64		Shape: productWidth
Description: Y coordinates in specified projection		
	standard_name	projection_y_coordinate
	long_name	y coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/algorithmCandidates/TSR/yCoordinateSpacing		
Type: Float64		Shape: NA
Description: Y coordinates spacing		
	long_name	Y coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/ancillaryData		
/science/LSAR/SME2/grids/ancillaryData/landCover		
Type: Int8		Shape: (productWidth, productLength)
Description: Land cover 01: Tree cover; 02: Shrublands; 03: Grassland; 04: Cropland; 05: Built-up; 06: Bare/sparse vegetation; 07: Snow & Ice; 08: Permanent water bodies; 09: Herbaceous wetland; 10: Mangroves; 11: Moss & lichen		
	long_name	Land cover
	_FillValue	-127
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/ancillaryData/localIncidenceAngle		
Type: Float32		Shape: (productWidth, productLength)
Description: Local incidence angle		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Local incidence angle
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: degrees		
/science/LSAR/SME2/grids/ancillaryData/localIncidenceAngleUncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: Local incidence angle standard deviation		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value

This document has been reviewed and determined not to contain export controlled technical data.

	sample_stddev	standard deviation of the dataset
	long_name	Local incidence angle standard deviation
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: degrees		
/science/LSAR/SME2/grids/ancillaryData/projection		
Type: Int32		
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	false_northing	0.0
	latitude_of_projection_origin	0.0
	longitude_of_projection_origin	0.0
	false_easting	0.0
	semi_major_axis	6378137
	inverse_flattening	298.25723
	longitude_of_central_meridian	0.0
	standard_parallel	30.0
	epsg_code	6933
	ellipsoid	WGS84
	grid_mapping_name	lambert_cylindrical_equal_area
	spatial_ref	PROJCS[WGS 84 / NSIDC EASE-Grid 2.0 Global,GEOGCS[WGS 84,DATUM[WGS_1984,SPHEROID[WGS 84,6378137,298.257223563,AUTHORITY[EPSG,7030]],AUTHORITY[EPSG,6326]],PRIMEM[Greenwich,0,AUTHORITY[EPSG,8901]],UNIT[degree,0.0174532925199433,AUTHORITY[EPSG,9122]],AUTHORITY[EPSG,4326]],PROJECTION[Cylindrical_Equal_Area],PARAMETER[standard_parallel_1,30],PARAMETER[central_meridian,0],PARAMETER[false_easting,0],PARAMETER[false_northing,0],UNIT[metre,1,AUTHORITY[EPSG,9001]],AXIS[Easting,EAST],AXIS[Northing,NORTH],AUTHORITY[EPSG,6933]]
/science/LSAR/SME2/grids/ancillaryData/surfaceQualityFlag		
Type: Int16		Shape: (productWidth, productLength)
Description: Surface quality flag Flag bits 0: Static water body flag; 1: Urban area flag; 2: Precipitation flag; 3: Snow or ice flag; 4: Permanent snow or ice flag; 5: Frozen ground flag based on soil surface temperature; 6: Terrain flag 7: Dense vegetation flag		
	long_name	Surface quality flag
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/ancillaryData/waterbodyFraction		
Type: Float32		Shape: (productWidth, productLength)

This document has been reviewed and determined not to contain export controlled technical data.

Description: Water body fraction		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Water body fraction
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/ancillaryData/xCoordinates		
Type: Float64		Shape: productLength
Description: X coordinates in specified projection		
	standard_name	projection_x_coordinate
	long_name	x coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/ancillaryData/xCoordinateSpacing		
Type: Float64		Shape: NA
Description: X coordinates spacing		
	long_name	X coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/ancillaryData/yCoordinates		
Type: Float64		Shape: productWidth
Description: Y coordinates in specified projection		
	standard_name	projection_y_coordinate
	long_name	y coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/ancillaryData/yCoordinateSpacing		
Type: Float64		Shape: NA
Description: Y coordinates spacing		
	long_name	Y coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/radarData		
/science/LSAR/SME2/grids/radarData/listOfFrequencies		
Type: List of string		Shape: N/A
Description: List of frequency layers available in the product		
	long_name	List of frequencies
/science/LSAR/SME2/grids/radarData/frequency[A|B]/centerFrequency		
Type: Float32		Shape: N/A
Description: Center frequency of the processed source data image in hertz		
	long_name	Center frequency

This document has been reviewed and determined not to contain export controlled technical data.

Unit: hertz		
/science/LSAR/SME2/grids/radarData/listOfPolarizations		
Type: List of string		Shape: N/A
Description: List of processed polarization layers with frequency [A|B]		
	long_name	List of polarizations
/science/LSAR/SME2/grids/radarData/frequency[A|B]/noiseEquivalentBackscatter[HH|HV|VH|VV]		
Type: Float32		Shape: (productWidth, productLength)
Description: Noise equivalent sigma nought of [HH|HV|VH|VV] polarization		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Noise equivalent sigma nought
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: decibel		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/numberOfLooks[HH|HV|VH|VV]		
Type: Float32		Shape: (productWidth, productLength)
Description: Number of looks within the 200m EASE-Grid 2.0 cell		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Number of looks
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/projection		
Type: Int32		
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	false_northing	0.0
	latitude_of_projection_origin	0.0
	longitude_of_projection_origin	0.0
	false_easting	0.0
	semi_major_axis	6378137
	inverse_flattening	298.25723
	longitude_of_central_meridian	0.0

This document has been reviewed and determined not to contain export controlled technical data.

	standard_parallel	30.0
	epsg_code	6933
	ellipsoid	WGS84
	grid_mapping_name	lambert_cylindrical_equal_area
	spatial_ref	PROJCS[WGS 84 / NSIDC EASE-Grid 2.0 Global,GEOGCS[WGS 84,DATUM[WGS_1984,SPHEROID[WGS 84,6378137,298.257223563,AUTHORITY[EPSG,7030]],AUTHORITY[EPSG,6326]],PRIMEM[Greenwich,0,AUTHORITY[EPSG,8901]],UNIT[degree,0.0174532925199433,AUTHORITY[EPSG,9122]],AUTHORITY[EPSG,4326]],PROJECTION[Cylindrical_Equal_Area],PARAMETER[standard_parallel_1,30],PARAMETER[central_meridian,0],PARAMETER[false_easting,0],PARAMETER[false_northing,0],UNIT[metre,1,AUTHORITY[EPSG,9001]],AXIS[Easting,EAST],AXIS[Northing,NORTH],AUTHORITY[EPSG,6933]]
/science/LSAR/SME2/grids/radarData/frequency[A|B]/rangeBandwidth		
Type: Float32		Shape: N/A
Description: Source data processed range bandwidth in hertz		
	long_name	Range bandwidth
Unit: hertz		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/sigma0[HH|HV|VH|VV]		
Type: Float32		Shape: (productWidth, productLength)
Description: Sigma nought at [HH|HV|VH|VV] polarization		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Sigma0 [HH|HV|VH|VV]
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/xCoordinates		
Type: Float64		Shape: productLength
Description: X coordinates in specified projection		
	standard_name	projection_x_coordinate
	long_name	x coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/xCoordinateSpacing		
Type: Float64		Shape: NA
Description: X coordinates spacing		
	long_name	X coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/yCoordinates		
Type: Float64		Shape: productWidth

This document has been reviewed and determined not to contain export controlled technical data.

Description: Y coordinates in specified projection		
	standard_name	projection_y_coordinate
	long_name	y coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/radarData/frequency[A|B]/yCoordinateSpacing		
Type: Float64		Shape: NA
Description: Y coordinates spacing		
	long_name	Y coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids		
/science/LSAR/SME2/grids/EASEGridColumnIndex		
Type: Int32		Shape: productLength
Description: EASE-Grid 2.0 column index		
	long_name	EASE-Grid 2.0 column index
Unit: unitless		
/science/LSAR/SME2/grids/EASEGridRowIndex		
Type: Int32		Shape: productWidth
Description: EASE-Grid 2.0 row index		
	long_name	EASE-Grid 2.0 row index
Unit: unitless		
/science/LSAR/SME2/grids/latitude		
Type: Float32		Shape: productWidth
Description: Latitude		
	long_name	Latitude
Unit: degree_north		
/science/LSAR/SME2/grids/longitude		
Type: Float32		Shape: productLength
Description: Longitude		
	long_name	Longitude
Unit: degree_east		
/science/LSAR/SME2/grids/projection		
Type: Int32		
Description: Product map grid projection: EPSG code, with additional projection information as HDF5 Attributes		
	false_northing	0.0
	latitude_of_projection_origin	0.0
	longitude_of_projection_origin	0.0
	false_easting	0.0
	semi_major_axis	6378137
	inverse_flattening	298.25723
	longitude_of_central_meridian	0.0

This document has been reviewed and determined not to contain export controlled technical data.

	standard_parallel	30.0
	epsg_code	6933
	ellipsoid	WGS84
	grid_mapping_name	lambert_cylindrical_equal_area
	spatial_ref	PROJCS[WGS 84 / NSIDC EASE-Grid 2.0 Global,GEOGCS[WGS 84,DATUM[WGS_1984,SPHEROID[WGS 84,6378137,298.257223563,AUTHORITY[EPSG,7030]],AUTHORITY[EPSG,6326]],PRIMEM[Greenwich,0,AUTHORITY[EPSG,8901]],UNIT[degree,0.0174532925199433,AUTHORITY[EPSG,9122]],AUTHORITY[EPSG,4326]],PROJECTION[Cylindrical_Equal_Area],PARAMETER[standard_parallel_1,30],PARAMETER[central_meridian,0],PARAMETER[false_easting,0],PARAMETER[false_northing,0],UNIT[metre,1,AUTHORITY[EPSG,9001]],AXIS[Easting,EAST],AXIS[Northing,NORTH],AUTHORITY[EPSG,6933]]
/science/LSAR/SME2/grids/retrievalQualityFlag		
Type: Int16		Shape: (productWidth, productLength)
Description: Soil moisture retrieval quality flag while 0 means clear and non-zero means anomaly Flag bits 0: Retrieval recommended flag; 1: Retrieval attempted flag; 2: Retrieval success flag; 3: Soil moisture thresholds between (0.02 and 0.60); 4: NISAR σ_{hh} anomaly flag; 5: NISAR σ_{hv} anomaly flag; 6: NISAR σ_{vv} anomaly flag; 7: NISAR σ_{vh} anomaly flag		
	long_name	Retrieval quality flag
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: unitless		
/science/LSAR/SME2/grids/soilMoisture		
Type: Float32		Shape: (productWidth, productLength)
Description: Soil moisture		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture
	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/soilMoistureUncertainty		
Type: Float32		Shape: (productWidth, productLength)
Description: Soil moisture uncertainty		
	min_value	minimum value
	max_value	maximum value
	mean_value	mean value
	sample_stddev	standard deviation of the dataset
	long_name	Soil moisture uncertainty

This document has been reviewed and determined not to contain export controlled technical data.

	_FillValue	-9999
	grid_mapping	projection
	_ChunkSize	chunk size of the dataset
Unit: meter^3/meter^3		
/science/LSAR/SME2/grids/xCoordinates		
Type: Float64		Shape: productLength
Description: X coordinates in specified projection		
	standard_name	projection_x_coordinate
	long_name	x coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/xCoordinateSpacing		
Type: Float64		Shape: NA
Description: X coordinates spacing		
	long_name	X coordinate of spacing
Unit: meters		
/science/LSAR/SME2/grids/yCoordinates		
Type: Float64		Shape: productWidth
Description: Y coordinates in specified projection		
	standard_name	projection_y_coordinate
	long_name	y coordinate of projection
Unit: meters		
/science/LSAR/SME2/grids/yCoordinateSpacing		
Type: Float64		Shape: NA
Description: Y coordinates spacing		
	long_name	Y coordinate of spacing
Unit: meters		
/science/LSAR/SME2/metadata/processingInformation		
algorithms Group		
/science/LSAR/SME2/metadata/processingInformation/algorithms/aggregation		
Type: String		Shape: NA
Description: Aggregation method applied to GCOV sigma0 to generate the 200m EASE-Grid 2.0 product		
	long_name	Aggregation method
/science/LSAR/SME2/metadata/processingInformation/algorithms/DSGAlgorithmReference		
Type: String		Shape: NA
Description: Reference to the DSG algorithm applied		
	long_name	Reference of the DSG algorithm
/science/LSAR/SME2/metadata/processingInformation/algorithms/PMIAlgorithmReference		
Type: String		Shape: NA
Description: Reference to the PMI algorithm applied		
	long_name	Reference of the PMI algorithm
/science/LSAR/SME2/metadata/processingInformation/algorithms/softwareVersion		

This document has been reviewed and determined not to contain export controlled technical data.

Type: String		Shape: NA
Description: Software version		
	long_name	Software version
/science/LSAR/SME2/metadata/processingInformation/algorithms/spatialFilter		
Type: String		Shape: NA
Description: Spatial filter method applied to GCOV sigma0 within a 200m window size		
	long_name	Spatial filter method
/science/LSAR/SME2/metadata/processingInformation/algorithms/TSRAlgorithmReference		
Type: String		Shape: NA
Description: Reference to the TSR algorithm applied		
	long_name	Reference of the TSR algorithm
inputs Group		
/science/LSAR/SME2/metadata/processingInformation/inputs/configFiles		
Type: List of string		Shape: NA
Description: List of input config files used		
	long_name	List of config files
/science/LSAR/SME2/metadata/processingInformation/inputs/l2GcovGranules		
Type: List of string		Shape: NA
Description: List of input L2 GCOV products used		
	long_name	List of input L2 GCOV granules
Parameters Group		
/science/LSAR/SME2/metadata/processingInformation/parameters/runConfigurationContents		
Type: String		Shape: NA
Description: Contents of the run configuration file with parameters used for processing		
	long_name	Contents of the runconfig file
/science/LSAR/SME2/metadata/sourceData		
/science/LSAR/SME2/metadata/sourceData/processingDateTime		
Type: String		Shape: NA
Description: Processing UTC date and time in the format YYYY-mm-ddTHH:MM:SS		
/science/LSAR/SME2/metadata/sourceData/productLevel		
Type: String		Shape: NA
Description: Product level. L0A: Unprocessed instrument data; L0B: Reformatted, unprocessed instrument data; L1: Processed instrument data in radar coordinates system; and L2: Processed instrument data in geocoded coordinates system		
/science/LSAR/SME2/metadata/sourceData/productVersion		
Type: String		Shape: NA
DescriptionProduct version which represents the structure of the product and the science content governed by the algorithm, input data, and processing parameters		
science/LSAR/SME2/metadata/sourceData/processingInformation		
Algorithms Group		
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/demInterpolation		
Type: String		Shape: NA

This document has been reviewed and determined not to contain export controlled technical data.

Description: DEM interpolation method	
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/geocoding	
Type: String	Shape: NA
Description: Geocoding algorithm	
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/geocodingAlgorithmReference	
Type: String	Shape: NA
Description: Reference to the geocoding algorithm applied (if applicable)	
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/polarimetricSymmetrization	
Type: String	Shape: NA
Description: Polarimetric symmetrization algorithm	
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/radiometricTerrainCorrection	
Type: String	Shape: NA
Description: Radiometric terrain correction (RTC) algorithm	
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/radiometricTerrainCorrectionAlgorithmReference	
Type: String	Shape: NA
Description: Reference to the radiometric terrain correction (RTC) algorithm applied (if applicable)	
/science/LSAR/SME2/metadata/sourceData/processingInformation/algorithms/softwareVersion	
Type: String	Shape: NA
Description: Software version used for processing	

This document has been reviewed and determined not to contain export controlled technical data.

APPENDIX A: ACRONYMS

ADT	Algorithm Development Team
ANF	Area Normalization Factor
AT	Along Track
ATBD	Algorithm Theoretical Basis Document
AWS	Amazon Web Services
BFPQ	Block (adaptive) Floating-Point Quantization (adaptive may indicate implementation options)
Cal/Val	Calibration and Validation (also sometimes cal/val)
CDR	Critical Design Review
CF	Climate and Forecast
CPU	Central Processing Unit
CRSD	Calibration Raw Signal Data
CSV	Comma-separated values
DAAC	Distributed Active Archive Center
DBF	Digital Beam Forming
DEM	Digital Elevation Model
DM	Diagnostic Mode
DN	Digital Number
DSG	Disaggregation
EAR	Export Administration Regulations
EASE	Equal-Area Scalable Earth
ECMWF	European Centre for Medium-Range Weather Forecasts
ECEF	Earth Centered Earth Fixed
ER#.#	Engineering Release #.#
ERA5	ECMWF Reanalysis 5th generation
FFT	Fast Fourier Transform
FM	Frequency Modulation
FOE	Forecast Orbit Ephemeris
FOV	Field of View
GCOV	Geocoded Polarimetric Covariance (L2_GCOV)
GCP	Ground Control Point
GDAL	Geospatial Data Abstraction Library
GDS	Ground Data System
GeoTIFF	Geographic Tagged Image File Format
GIS	Geographic Information System
GMTED	Global Multi-resolution Terrain Elevation Data
GNSS	Global Navigation Satellite System
GOFF	Geocoded Pixel Offsets (L2_GOFF)
GPU	Graphics Processing Unit
GSLC	Geocoded Single Look Complex (L2_GSLC)
GUNW	Geocoded Unwrapped Interferogram (L2_GUNW)
HH	Horizontal-transmit, Horizontal-receive polarization
HK, HKTM	Housekeeping Telemetry
HDF5	Hierarchical Data Format version 5

This document has been reviewed and determined not to contain export controlled technical data.

HV	Horizontal-transmit, Vertical-receive polarization
ICU	Integrated Correlation Unit
InSAR	Interferometric Synthetic Aperture Radar
ISCE	InSAR Scientific Computing Environment
ISCE3	InSAR Scientific Computing Environment Enhanced Edition (for NISAR)
ISO	International Organization for Standardization
ISRO	Indian Space Research Organisation (British spelling)
JPL	Jet Propulsion Laboratory
JSON	JavaScript Notation
L0B	Level-0B (data)
L1	Level-1 (data)
L2	Level-2 (data)
L3	Level-3 (data)
LRR	[JPL] Limited Release Request
LRS	[JPL] Limited Release System
LUT	Lookup Table
Mbps	Megabits per second
MHz	Megahertz
MOE	Medium-precision Orbit Ephemeris
NASA	National Aeronautics and Space Administration
NETCDF4	Network Common Data Format 4 (also netCDF4)
NISAR	NASA-ISRO Synthetic Aperture Radar
NOE	Near-Realtime Orbit Ephemeris
OpenMP	Open Multi-Processing
PCM	Process Control Management
PDF	Portable Document Format (often pdf)
PDR	Preliminary Design Review
PMI	Physical Model Inversion
POD	Precision Orbit Determination
POE	Precision Orbit Ephemeris
PRF	Pulse Repetition Frequency
QA	Quality Assurance
R#.#	Release #.# (.0 often not used)
REE	Radar Echo Emulator
RFI	Radio Frequency Interference
RIFG	Range-Doppler Interferogram (L1_RIFG)
RMS	Root Mean Square
RMSE	Root Mean Square Error
ROFF	Range-Doppler Pixel Offsets (L1_ROFF)
RRSD	Raw Radar Signal Data
RRST	Raw Radar Signal Telemetry
RSLC	Range-Doppler Single Look Complex (L1_RSLC)
RTC	Radiometric Terrain Correction
RUNW	Range-Doppler UnWrapped Interferogram (L1_RUNW)
RV	Right-circular, V-receive compact polarization
SAR	Synthetic Aperture Radar (L-SAR: L-band. S-SAR: S-band)

This document has been reviewed and determined not to contain export controlled technical data.

SAS	Science Algorithm Software
SDS	Science Data System
SDT	Science Definition Team
SIS	Software Interface Specification
SLC	Single Look Complex
SME2	Soil Moisture product based on a 200-meter global EASE Grid projection
SMAP	Soil Moisture Active Passive (Mission)
SMOS	Soil Moisture and Ocean Salinity
SNAPHU	Statistical-cost, Network-flow Algorithm for Phase Unwrapping
SWST	Sampling Window Start Time
SRTM	Shuttle Radar Topography Mission
ST	Science Team
SWST	Sampling Window Start Time
TAI	International Atomic Time (Temps Atomique International)
TCF	Terrain Correction Factor
TEC	Total Electron Content
TFdb	Trackframe Database
TSR	Time Series Ratio
UR	Urgent Response
UTC	Universal Time Coordinated
UTM	Universal Transverse Mercator
VH	Vertical-transmit, Horizontal-receive polarization
VV	Vertical-transmit, Vertical-receive polarization
VWC	Vegetation Water Content
WGS84	World Geodetic System 84
XML	eXtensible Markup Language (xml in code)
YAML	YAML Ain’t Markup Language

This document has been reviewed and determined not to contain export controlled technical data.